

Math 104 - Spring 2026 - Exploration 3.4B: Reese's Pieces

Reese's Pieces candies come in three colors: orange, yellow, and brown. Suppose you take a random sample of candies and want to estimate the long-run proportion of orange candies.

- (1) Assume that the true long-run proportion of orange candies is $\pi = 0.50$. If you take a random sample of 100 Reese's Pieces and find the sample proportion, is it *guaranteed* to equal 0.50?
- (2) Assuming $\pi = 0.50$, if we calculate a confidence interval based on a sample proportion, is it guaranteed that the interval will contain the value 0.50?
- (3) Still assuming $\pi = 0.50$, if you select another random sample of 100 Reese's Pieces, is it guaranteed that the second sample proportion will be the same as the first?

Will the confidence interval based on the new sample be the same as the confidence interval based on the first sample?

What aspect(s) of the interval would differ, if any?

We're going to explore the behavior of confidence intervals from different random samples. Open the **Simulating Confidence Intervals** applet.

- Make sure that **Proportions**, **Binomial**, and **Wald** are selected from the drop-down menus.
- Make sure that π is set to be 0.50 and **n** is 100.
- Set **Intervals** to 1.
- Make sure **Conf level** is set to 95%.
- Press **Sample**.

Notice that a horizontal line is added to the graph in the middle. Your line may be red or green. This line represents a confidence interval. The dot in the middle of the line corresponds to the observed sample proportion. The interval is drawn relative to the horizontal axis at the bottom of the figure. The vertical line labeled 0.500 represents the parameter. You can also see the sample proportion plotted in the graph labeled Sample Statistics.

- (4) Click on the interval. This reveals the value of the midpoint $\hat{p}$ (at the bottom of the middle graph) and the endpoints of the 95% confidence interval. Record the following.
 - (a) The sample proportion $\hat{p}$.
 - (b) The 95% confidence interval.
 - (c) Is the sample proportion within the confidence interval? Why is this unsurprising?
- (d) Is the long-run proportion $\pi = 0.50$ within the confidence interval?

Notice that 0.50 is either in the interval you recorded or it isn't. If you recorded an interval like (0.473, 0.537), it's not the case that 0.50 is sometimes in that interval and sometimes it isn't; it's in the interval, period. This is true, even when you don't know the long-run proportion π . Thus it is **incorrect** to say "There's a 95% chance that π is between 0.473 and 0.537." So the big question is:

What is happening 95% of the time?

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Now change **Intervals** from 1 to 99 for a total of 100 random samples from a process where $\pi = 0.50$. Press **Sample**. This produces a total of 100 sample proportions (graphed in the Sample Statistics graph) and 100 confidence intervals (at the 95% confidence level, shown in the middle graph) based on those sample proportions.

- (5) Study the graph labeled **Sample Statistics** closely. At about what number is this graph of sample proportions centered? Is this surprising? Why or why not?
- (6) Notice that some of the dots on the **Sample Statistics** graph are colored red and some are green. Click on a **red** dot. Doing this will reveal the value of the sample proportion and the corresponding 95% confidence interval. Record the following.
 - (a) The sample proportion.
 - (b) The 95% confidence interval.
 - (c) Is the true process probability ($\pi = 0.50$) within the confidence interval?
- (7) Now click on a **green** dot and repeat the previous four items.
 - (a) The sample proportion.
 - (b) The 95% confidence interval.
 - (c) Is the true process probability ($\pi = 0.50$) within the confidence interval?

The graph in the middle displays the 100 different 95% confidence intervals, depicting in green the ones that succeed in capturing the actual value of the long-run proportion and in red the intervals that fail to capture the actual value of π .

- (8) In the bottom left corner, the applet maintains a running total for the percentage of the intervals produced so far that contain $\pi = 0.50$. What percentage of the 100 random intervals you have generated succeeded in capturing the value of the 0.50 inside the interval?
- (9) Press the **Sample** a few times and watch how, if at all, the percentage reported under Running total changes. Based on your observations, fill in the blanks:

95% confidence means that if we repeatedly sampled a random process, and used the sample statistic to construct a 95% confidence interval each time, in the long run, roughly % of the intervals constructed would contain the value of the parameter, and the remaining % would not.

- (10) Now we'll consider the effects of changing the confidence level and the sample size.
 - (a) Change **Conflevel** to 99% and press **Recalculate**.
 - (i) How do the widths of the intervals change? Why does this make sense?
 - (ii) How does the running total change? Why does this make sense?
 - (b) Now change **n** to 400 and press **Sample**.
 - (i) How do the widths of the intervals change? Why does this make sense?
 - (ii) How does the running total change? Why does this make sense?